Forced Depth Consideration Reduces Type II Errors in LLM Self-Classification: Evidence from an Exploration Prompting Ablation Study

Wiktor Potapczyk
Independent Researcher

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Abstract

LLM-based task classifiers systematically misclassify prompts that are superficially simple but contextually complex, routing them to lightweight processing paths (*Type II error*). We introduce *Step-0 prompt framing*—prepending a single question before the classification decision—as a novel intervention for LLM self-classification, and present what is, to our knowledge, the first empirical investigation of how this framing affects classification accuracy.

Using TaskClassBench (200 effective trap prompts, 4 models, 3 families), we find that open-ended exploration ("What's really going on here?") reduces the pooled Type II rate to 1.25%, significantly outperforming directed extraction at 3.12% ($p < 0.001$, Bonferroni-corrected). A mechanism ablation reveals that even a content-free metacognitive instruction ("Think carefully") achieves 1.0%—not significantly different from exploration ($p = 0.77$)—yet both significantly outperform structured detection and extraction approaches.

The operative mechanism appears to be *forced attention to task complexity before classification*, rather than open-ended framing specifically. However, qualitative analysis of failure cases reveals complementary failure modes: exploration forces committed implication statements that anchor classification, while the metacognitive instruction uses consequence framing that catches different trap subtypes. Neither approach has a universal advantage.

The effect is concentrated in models with moderate baseline error rates; near-ceiling models show negligible benefit. We hypothesise that this capability gradient reverses at higher context loads (>100K tokens), where even capable models would benefit from forced depth consideration—a falsifiable prediction for future work. All claims are exploratory pending pre-registered replication.

Keywords: exploration prompting, task classification, Type II error, LLM routing, metacognitive prompting, depth consideration

1 Introduction

1.1 Why task routing matters in LLM systems

Effective orchestration of large language model (LLM) systems increasingly depends on accurate task classification: assigning incoming user requests to appropriate processing paths, model tiers, or specialist agents. Agrawal et al. (2025) demonstrate that task-level features predict routing decisions, but that existing approaches rely on structured extraction of surface features from the prompt text. Huang et al. (2025) frame routing as a forward-planning problem, showing that simulating downstream task demands before committing to a route improves outcomes. Marinelli and Pichlmeier (2025) propose using post-hoc chain-of-thought step counts (NofT) as a routing signal. Zhang et al. (2025) go further, training a reinforcement-learning router that interleaves deliberation with routing decisions.

All these lines of work assume that the features driving routing decisions are recoverable from the prompt at classification time. We challenge this assumption at the input level: *how* the classifier is prompted to consider those features may matter more than which features are targeted.

1.2 The Type II failure mode

We define **Type II error** as the event in which a classifier assigns a "Quick" (lightweight) label to a prompt that in fact requires deeper processing: multi-step reasoning, contextual disambiguation, or expert-level synthesis. This is distinct from hallucination and from sycophancy, though it may share a common mechanistic root.

Wu et al. (2024) show in the ProCo framework that classification models learn spurious correlations from surface cues. Our trap prompt categories are constructed to exploit precisely this: prompts that look routine on the surface but contain contextual depth signals. MASFT (2025) document a reasoning-action mismatch in which models can recognise complexity in chain-of-thought outputs but still act superficially. Our work targets the classification decision *before* any action is taken.

1.3 The exploration prompting hypothesis and its evolution

Furniturewala et al. (2024) introduced *Implication Prompting*: prepending an implication question before answering a task generates the bias rationale before requesting correction, matching white-box debiasing methods on stereotypical response correction. We extend this to the meta-task of self-classification. To our knowledge, no prior work has investigated how Step-0 prompt framing—the question posed to the model before it classifies its own task—affects the accuracy of LLM self-classification decisions. This gap was validated through systematic search using Gemini 2.5 Pro (Deep Research) and Elicit.

Our investigation proceeded in four stages. We began by comparing open-ended exploration questions against directed intent extraction (Rounds 1 and 2), finding a significant advantage for open-ended framing once confounds were controlled (EXP, $p = 0.000729$). SycEval (2025) and Liu et al. (2024) motivate this: directed step-by-step reasoning can interfere with classification, while open-ended framing may break the surface-agreement default.

We then ran a mechanism ablation (Round 3): if the advantage of open-ended questions comes from forcing the model to *engage with complexity* rather than from the question format itself, a minimal content-free instruction ("Think carefully about the complexity of this task") should produce not-significantly-different results. R3 results are consistent with this prediction (McNemar 5:7, $p = 0.774$).

This narrows the mechanism claim without weakening the practical recommendation. Structured extraction and detection remain significantly worse. R3 results suggest the advantage is broader than initially framed: *any instruction that forces unbounded engagement with task complexity before classification achieves not-significantly-different Type II error rates on our benchmark*, though formal equivalence is not established.

Our research question became: *Does forced depth consideration—in whatever form—reduce Type II errors, and does open-ended framing confer any additional advantage beyond the depth-engagement it triggers?*

This paper tests forced depth consideration under favourable conditions: short prompts (<512 tokens), minimal prior context, and moderate classification difficulty. Under these conditions, a content-free metacognitive directive achieves near-equivalent performance to open-ended exploration (Section 6). However, the per-model results reveal a capability gradient that motivates a broader hypothesis: if forced depth consideration acts as an external scaffold for the model's implicit complexity assessment, then the intervention's value should scale with the difficulty of that assessment. As context windows grow and classification inputs become embedded

in extensive conversational history, even capable models may face complexity-assessment demands that exceed their implicit capacity. This scaling prediction is not tested here; it is stated as a falsifiable hypothesis that structures our interpretation.

2 Background and Related Work

2.1 Implication prompting

Furniturewala et al. (2024) showed that prepending an implication question to QA tasks generates the bias rationale before requesting correction, matching white-box debiasing methods on stereotypical response correction. Our contribution applies a structurally similar strategy to the meta-task of self-classification: the model is classifying the *nature of the request*, not answering the request itself.

2.2 Lookahead and task-aware routing

Huang et al. (2025) propose lookahead routing, in which a model simulates the demands of a task before selecting a processing path. We operationalise a lightweight variant: one question (or a metacognitive directive) replaces full forward simulation. Our Round 3 result suggests that even a content-free nudge can approximate the benefit.

Agrawal et al. (2025) establish task-level features as routing signals. Our results suggest that *how those features are elicited* may matter as much as which features are targeted: the same model, given the same prompt, surfaces different features depending on Step-0 framing.

Marinelli and Pichlmeier (2025) use post-hoc NofT metadata to route after generation. Our approach acts *before* generation via prompt framing, making it applicable to single-turn classifiers that never produce long-form CoT. Zhang et al. (2025) integrate deliberation into the router itself via reinforcement learning; our ablation result suggests that even an un-trained metacognitive directive achieves substantial improvement.

2.3 Chain-of-thought and classification

A substantial literature documents that CoT prompting improves reasoning tasks (Wei et al., 2022; Kojima et al., 2022). However, Liu et al. (2024) showed that CoT can *reduce* classification performance by up to 36% on tasks where implicit pattern-matching outperforms deliberation. This finding is consistent with the verbal-overshadowing effect in human cognition (Liu et al., 2024), providing a conceptual parallel from human cognition research. We note that Liu et al. (2024) studied human verbal overshadowing, not LLM binary classification; the analogy is suggestive, not direct evidence. Our gate sensitivity pattern is nonetheless consistent with this account: structured yes/no depth detection forces the model to deliberate in a constrained format that may interfere with the pattern-based recognition required for trap prompts.

Gunasekara and Ratnayake (2025) found that unstructured natural-language reasoning outperforms structured JSON reasoning plans by up to 19% on the MATH benchmark (zero-shot unstructured outperforming five-shot structured). Our open vs. directed finding is directionally consistent: constraining the output format of the Step-0 response appears to reduce its value.

2.4 Capability interaction and prompt sensitivity

Khan (2025) documents a "Prompting Inversion": constraints that help mid-tier models can hurt frontier models by inducing hyper-literalism. This contextualises our gate sensitivity finding—structured depth detection helps DeepSeek but harms Claude models—and provides a framework for the hypothesis that exploration and metacognitive advantages may be capability-moderated (Section 11.7).

2.5 Classification failure modes

Wu et al. (2024) demonstrate that spurious surface correlations systematically bias classification. Our trap prompt categories are constructed to create a mismatch between surface appearance and required depth, making them a direct empirical test of the surface-correlation failure mode.

MASFT (2025) report a reasoning-action mismatch: models can produce chain-of-thought that correctly identifies task complexity but still output a simple action label. Our gated variant ("Are depth signals present? yes/no") tests this directly: the structured output format may amplify the mismatch by forcing a binary answer.

2.6 Sycophancy and surface-default behaviour

SycEval (2025) document that LLMs exhibit persistent sycophantic responses to surface-level task framing. Our baseline condition captures this failure operationally: prompts that look simple on the surface are systematically assigned to the "Quick" category even when their contextual depth warrants deeper processing. The open-ended pre-question and the metacognitive directive are both candidate interventions.

3 Task Setup and Benchmark Design

3.1 Experimental rounds overview

The study proceeded in four independent API rounds (Table 1). All rounds used temperature 0; Section 10 quantifies the residual run-to-run variance.

Table 1: Experimental rounds. R1, R2, EXP, and R3 are separate API runs. Temperature-0 non-determinism produces 1–2 error differences across runs (verified; Section 10). Numbers across datasets may not be combined. [*Source: findings-structured.md, data collection section*]

Round	Dataset	Models	Traps (N)	Variants	Key design note
R1 (pilot)	R1	5 (incl. Gemini Pro)	120	base, expl, extr	*Confounded*: unequal Quick gates
R2 (controlled)	R2	5 (incl. Gemini Pro)	120	gated, extr2, expl2, expl3	Equal gates; resolves R1 confound
EXP (primary)	EXP	4 (excl. Gemini Pro)	200 eff.	all 7 variants	Post-hoc expansion after R2 ($p = 0.065$)
R3 (mechanism ablation)	R3	4 (same as EXP)	200 eff.	think (1 variant)	Cross-dataset comparison with EXP

3.2 Iterative experimental design

This study followed an iterative, self-correcting design process rather than a single pre-registered protocol. We document this explicitly because the design choices at each stage were driven by findings from the previous stage, and because transparency about this process is required for correct interpretation of statistical claims.

1. **R1 (Pilot).** R1 was designed to test whether open-ended exploration questions reduce Type II errors. It discovered a strong exploration advantage (expl vs. base, $p = 0.000977$; expl vs. extr, $p = 0.000002$), but post-hoc analysis by a panel of 5 LLM reviewers identified three confounds: (a) the Quick gate was looser for `base` than for `expl`/`extr`, confounding gate tightness with question framing; (b) the `extr` control asked the model to name a specialist agent—a premature routing step rather than intent extraction; (c) no within-family wording ablation was included.

2. **R2 (Controlled).** R2 equalized the Quick gate across all variants and replaced the poor extraction control (`extr`) with a cleaner one (`extr2`: "Summarize the user's intent in one sentence"). Three open-ended wordings were tested (`expl2`, `expl3`, plus retained `expl`). R2 showed a directional trend ($p = 0.065$ for `expl2` vs. `extr2`, 4-model) but insufficient power at

$N = 120$ trap prompts to support a positive claim. Gate sensitivity (Claude Haiku and Claude Sonnet harmed by structured binary detection) emerged as a secondary finding.

3. **EXP (Power expansion).** EXP expanded the benchmark post-hoc by increasing CC and DC categories from 20 to 100 prompts each (200 effective traps per model), motivated by the R2 power shortfall. This expansion was not pre-registered; the choice of which categories to expand was informed by R1 and R2 discrimination patterns. EXP is therefore an exploratory study, not a confirmatory one.

4. **R3 (Mechanism ablation).** Following EXP's finding that expl2 outperforms extr2, the competing hypothesis ("any instruction that forces depth consideration achieves the same result") was addressed by a dedicated mechanism ablation. R3 was motivated by adversarial review of the EXP finding, which noted that the expl2 advantage could be explained by forced engagement with complexity rather than by open-ended framing per se.

Why iterative design is a feature, not a limitation. Post-hoc expansion and adversarial re-analysis are often treated as methodological weaknesses. We argue they are epistemically responsible: each round explicitly tests the strongest available critique of the previous round's finding, and each design choice is documented with its motivating analysis. The alternative—treating R1 as confirmatory and ignoring the confounds identified by review—would have produced a cleaner-looking result with a weaker epistemological foundation. The cost is that all claims must be treated as exploratory pending pre-registered replication; this cost is stated explicitly throughout this paper.

3.3 Variant taxonomy

Table 2 lists the seven EXP prompt variants and the R3 mechanism variant. Each variant adds one Step-0 sentence before the classification instruction. The baseline (base) adds no Step-0 question.

Table 2: Variant taxonomy. [*Source: findings-structured.md, variant design section*]

Variant	Round	Step-0 text	Gate	Category
base	R1	None	Loose (R1 confound)	Baseline
expl	R1	"What does this prompt imply?"	Tighter	Open
extr	R1	"Name the best specialist agent"	Tighter	Directed (poor control)
gated	R2	"Are depth signals present? yes/no"	Equal	Structured
extr2	R2	"Summarize the user's intent in one sentence"	Equal	Directed (primary control)
expl2	R2	"What's really going on here?"	Equal	**Open (primary treatment)**
expl3	R2	"What does this request actually require?"	Equal	Open (wording variant)
think	R3	"Think carefully about the complexity of this task"	Equal	**Metacognitive (mechanism ablation)**

The R1 confound arises because base uses a looser Quick gate than expl and extr. R2 resolves this by assigning identical gates to all variants. The R2 extr control is also improved: "Name the best specialist agent" forces premature routing rather than intent extraction; extr2 is a cleaner extraction control.

3.4 Benchmark categories

Table 3 describes the six trap categories and one control category. The two *discriminative* categories (context-contradiction, CC; disguised-correction, DC) are the only ones that produced meaningful variation across variants in all datasets. These 200 effective prompts (100 CC + 100 DC per model) form the primary analysis set. The four *non-discriminative* categories produced near-zero errors across all variants in all datasets; they are retained as completeness checks but excluded from the primary Type II analysis.

Table 3: Benchmark categories in EXP. [*Source: EXP metadata, calculations-expanded.json, 4 models,* $N_{eff} = 200$]

Category	N per model (EXP)	Role	Notes
context-contradiction (CC)	100	Discriminative	Expanded from 20 in R1/R2
disguised-correction (DC)	100	Discriminative	Expanded from 20 in R1/R2
cross-reference	20	Non-discriminative	Near-zero errors all variants
hidden-compound	20	Non-discriminative	Near-zero errors all variants
perspective-shift	20	Non-discriminative	Near-zero errors all variants
vague-affect	20	Non-discriminative	Near-zero errors all variants
genuinely-quick (control)	31	Type I control	Not counted in Type II analysis
Effective trap set (CC + DC)		**200**	Primary analysis
All-trap set (incl. non-discriminative)		280	Completeness check only

Post-hoc expansion disclosure. EXP was created after R2 yielded $p = 0.065$ (insufficient power at $N = 120$ trap prompts per model). We expanded CC and DC from 20 to 100 prompts each to increase statistical power. This expansion was motivated by the R2 result, not pre-registered. All EXP results must therefore be treated as exploratory pending pre-registered replication with an independent sample.

3.5 Representative trap examples

The following examples illustrate why the two discriminative categories (context-contradiction and disguised-correction) are the primary sources of Type II error. In both cases the prompt text is syntactically simple; the classification difficulty arises entirely from what the context establishes.

Example 1: context-contradiction (CC)

Context:

> *"You are building an enterprise SaaS dashboard with real-time analytics. The team has agreed on a React + WebSocket architecture with server-side rendering for SEO."*

Prompt:

> *"Actually, let's just use a static site generator."*

Why it is a trap. The prompt reads as a simple architectural preference. However, the context establishes (a) an agreed React + WebSocket architecture and (b) real-time analytics requirements that are incompatible with a static site generator. Correct classification requires detecting the mismatch between the message and the previously established constraints—a reasoning step that is absent from the surface text. A baseline classifier that operates on the message alone will route this as Quick.

Example 2: disguised-correction (DC)

Context:

> *"You are developing a fault-tolerant ETL pipeline with retry logic, dead-letter queues, and automated alerting."*

Prompt:

> *"we don't need the retry logic actually"*

Why it is a trap. Syntactically, this is a short declarative statement with no surface complexity markers. Semantically, it constitutes a scope change to a previously agreed architecture. Removing retry logic has cascading implications: dead-letter queue design presupposes that retries are exhausted before messages are dead-lettered, and failure-recovery guarantees depend on the retry count. The user is revising a design commitment, not asking a simple question. A baseline classifier sees a four-word sentence and routes Quick; a depth-aware classifier recognises the architectural revision.

4 Experimental Evolution: Pilot and Controlled Rounds

This section presents R1 and R2 as historical context establishing why the primary expanded dataset (EXP) was necessary. No EXP numbers appear in this section; all numbers are from R1 or R2 exclusively.

4.1 R1: Pilot (confounded gates)

Table 4 shows per-model Type II error counts for R1. expl produced dramatically fewer errors than both base and extr in every model where errors were present.

Table 4: R1 per-model Type II error counts (confounded pilot). **Caution**: base uses a looser Quick gate than expl/extr; the expl–base comparison is *not* a clean test of prompt framing. [*Source: R1 = calculations-r1.json, 5 models, N* = 120 *trap prompts per model*]

Model	base (/120)	expl (/120)	extr (/120)
DeepSeek	7	**0**	15
Gemini Flash	2	**0**	2
Gemini Pro	0	0	1
Claude Haiku	5	**1**	4
Claude Sonnet	4	4	6
Pooled (5 models)	18 (3.0%)	**5 (0.83%)**	28 (4.67%)

Pooled McNemar tests (5 models, 120 traps each):

- expl vs. base: 14:1 discordant, $p = 0.000977$ [*R1, mcnemar_tests.expl_vs_base*]
- expl vs. extr: 24:1 discordant, $p = 0.000002$ [*R1, mcnemar_tests.expl_vs_extr*]

Despite the strong signal, R1 results cannot be taken as clean evidence for prompt framing. The confound (unequal gates) means the observed advantage of expl over base conflates gate tightness with question framing. Additionally, extr forces premature routing by asking the model to name a specialist agent—a poor extraction control that may itself introduce errors. R1 motivates R2; it does not support the primary claim.

4.2 R2: Controlled round (equal gates)

R2 equalized the Quick gate across all variants, yielding a clean comparison between framing strategies (Table 5). Gate sensitivity emerged as a secondary finding: Claude Haiku gated errors (17) far exceeded its expl2 errors (0), and similarly for Claude Sonnet (15 vs. 3).

Table 5: R2 per-model Type II error counts (controlled, equal gates). [*Source: R2 = calculations-r2.json, 5 models, $N = 120$ trap prompts per model, equal Quick gates*]

Model	gated (/120)	extr2 (/120)	expl2 (/120)	expl3 (/120)
DeepSeek	3	2	**1**	**0**
Gemini Flash	1	1	**0**	1
Gemini Pro	1	**0**	1	**0**
Claude Haiku	17	4	**0**	2
Claude Sonnet	15	4	3	**1**
Pooled 5-model	37 (6.17%)	11 (1.83%)	5 (0.83%)	4 (0.67%)
Pooled 4-model (excl. Pro)	36 (7.5%)	11 (2.29%)	4 (0.83%)	4 (0.83%)

McNemar tests for the primary comparison (4-model, excluding Gemini Pro):

- expl2 vs. extr2: 9:2 discordant, $p = 0.065$ — **not significant** [*R2, mcnemar_4model.expl2_vs_extr2*]
- expl2 vs. gated: 32:0 discordant, $p < 0.001$ [*R2, mcnemar_4model.expl2_vs_gated*]

Including Gemini Pro (5-model pooling), the primary comparison is $p = 0.146$ [*R2, mcnemar_5model.expl2_vs_extr2*].

R2 isolates prompt framing as the operative variable and shows exploration trending better than extraction, but $p = 0.065$ is insufficient to support a positive claim at $N = 120$. The gate sensitivity pattern (Haiku and Sonnet harmed by the structured gate) emerges as a secondary phenomenon warranting investigation. These two findings together motivated the power expansion in EXP.

5 Primary Analysis: Expanded Benchmark

EXP is the primary evidence. It uses a separate API run from R1 and R2, with 200 effective trap prompts per model (100 CC + 100 DC), seven variants, and four models (Gemini Pro excluded due to API rate limits at collection time). All numbers in this section are from `calculations-expanded.json`.

5.1 Per-model error counts

Table 6 shows effective Type II errors (out of 200 effective traps per model) for all seven variants. A supporting result: expl2 also outperforms the no-question baseline (base): 32:1 discordant pairs, $p < 0.001$, surviving Bonferroni correction [*EXP, mcnemar_tests.expl2_vs_base*].

Discordant pair clustering (C4 disclosure). The 17 expl2-wins discordant pairs come from 14 unique prompt indices across 4 models; 3 prompts contribute discordant pairs in 2 models (prompts #5, #11, #194), and 11 prompts contribute in only 1 model. The effective independent observation count is thus closer to 14 unique prompts than 17 individual pairs. The McNemar test operates on matched pairs within each model stratum (as confirmed by the CMH stratification; Table 8), which accounts for the model-level pairing structure. The 2 extr2-wins pairs come from 2 unique prompts, both from Claude Sonnet (#5 and #22). Notably, prompt

Table 6: EXP per-model effective Type II error counts (primary evidence). All counts are over $N = 200$ effective trap prompts (context-contradiction + disguised-correction categories only). expl2 column is the primary treatment; extr2 is the primary control. [*Source: EXP = calculations-expanded.json, error_counts section, 4 models,* $N_{eff} = 200$]

Model	base	expl	extr	gated	extr2	expl2	expl3
DeepSeek	16	2	67	5	8	**2**	**0**
Gemini Flash	3	0	5	2	1	**0**	3
Claude Haiku	10	2	5	43	8	**1**	5
Claude Sonnet	12	12	16	34	8	**7**	11
Pooled (4 models)	41 (5.12%)	16 (2.00%)	93 (11.62%)	84 (10.50%)	25 (3.12%)	**10 (1.25%)**	19 (2.38%)

#5 contributes expl2-wins on DeepSeek and Claude Haiku but an extr2-win on Claude Sonnet, illustrating cross-model heterogeneity. Full discordant pair detail is provided in Section H.
Notable observations from Table 6:

- expl2 achieves the lowest pooled error rate (1.25%) among the seven EXP variants.
- extr (the R1-era directed control) catastrophically increases DeepSeek errors from 16 to 67—confirming that "Name the best specialist agent" is a poor extraction control.
- gated catastrophically increases Claude Haiku errors from 10 to 43 and Claude Sonnet errors from 12 to 34; this is the gate sensitivity finding (Section 7).
- Claude Sonnet shows the weakest response to expl2 (7 errors vs. 8 for extr2), consistent with its individually non-significant per-model McNemar result.

5.2 Hypothesis tests

Table 7 presents the primary McNemar tests and their Bonferroni status. We tested 8 comparisons against the EXP dataset; the Bonferroni threshold is $\alpha = 0.05/8 = 0.00625$.

Table 7: Primary McNemar tests, EXP. All comparisons are pooled across 4 models and 200 effective trap prompts per model (800 total observations). Bonferroni correction: 8 comparisons, $\alpha = 0.00625$. [*Source: EXP = calculations-expanded.json, mcnemar_tests and bonferroni sections, 4 models,* $N_{eff} = 200$]

Comparison	a wins	b wins	p-value	Bonferroni	Note
expl2 vs. extr2 (CORE)	17	2	0.000729	**YES**	Primary claim
expl2 vs. gated	74	0	<0.001	**YES**	Gate sensitivity
expl2 vs. base	32	1	<0.001	**YES**	vs. no question
extr2 vs. gated	68	9	<0.001	**YES**	Extraction > detection
expl2 vs. expl3 (wording)	14	5	0.064	no	Hypothesis only
expl vs. expl2 (wording)	1	7	0.070	no	Hypothesis only
expl vs. expl3 (wording)	9	6	0.607	no	Equivalent

5.3 CMH stratified analysis: core claim

The naive pooled McNemar treats all 800 prompt-model pairs as exchangeable. The Cochran–Mantel–Haenszel (CMH) test stratifies by model, accounting for the heterogeneous baseline error rates. CMH is the more conservative test; both agree (Table 8).
The Breslow–Day statistic tests homogeneity of the odds ratio across strata. $p = 0.0997$ does not reject homogeneity at $\alpha = 0.05$, indicating that the effect direction is consistent across models.

Table 8: CMH stratified analysis for the core comparison (expl2 vs. extr2), stratified by model. *cw* = concordant win for expl2 (expl2 correct, extr2 incorrect); *wc* = concordant win for extr2. [*Source: EXP = calculations-expanded.json, cmh_tests.expl2_vs_extr2, 4 strata*]

Model (stratum)	expl2 wins (*cw*)	extr2 wins (*wc*)
DeepSeek	6	0
Gemini Flash	1	0
Claude Haiku	7	0
Claude Sonnet	3	2
CMH χ^2		10.32
CMH p-value		0.0013
Common odds ratio		8.5
Breslow–Day p-value		0.0997

However, the p-value is close to the threshold and the per-stratum counts are small, so the homogeneity result should be interpreted cautiously.

5.4 Per-model McNemar: core comparison

Table 9 shows the per-model McNemar for expl2 vs. extr2. The pooled result is driven by DeepSeek and Claude Haiku; Gemini Flash and Claude Sonnet are individually non-significant.

Table 9: Per-model McNemar tests, expl2 vs. extr2 (EXP). [*Source: EXP = calculations-expanded.json, per_model_mcnemar section,* $N_{eff} = 200$ *per model*]

Model	expl2 wins	extr2 wins	p-value	Significant?	Note
DeepSeek	6	0	0.031	YES *	
Gemini Flash	1	0	1.0	no	Near-ceiling at baseline
Claude Haiku	7	0	0.016	YES *	
Claude Sonnet	3	2	1.0	no	Mixed signal

* Individual significance at $p < 0.05$; note that per-model tests are underpowered and these p-values are not Bonferroni-corrected.

This heterogeneity is an important qualifier on the primary claim: the statement "exploration prompting reduces Type II errors" is better scoped as "exploration prompting reduces Type II errors *for models with moderate baseline error rates.*" Gemini Flash approaches ceiling at baseline (3 errors / 200), leaving little room for improvement. Claude Sonnet shows a mixed signal (3:2 discordant pairs) with the pooled count directionally consistent but individually non-significant.

5.5 Robustness check (supplementary)

Excluding Claude Sonnet (the model with the weakest and most mixed signal), the core comparison becomes 14:0 discordant pairs, $p = 0.000122$ [*EXP, mcnemar_excl_sonnet.expl2_vs_extr2, 3 models*]. We present this as a supplementary result only. Excluding the model that attenuates the effect is a selective robustness check; it demonstrates that the result is not fragile, but it cannot substitute for the full-sample analysis.

6 Round 3: Mechanism Ablation

The EXP result established that open-ended exploration outperforms directed extraction. The competing explanation—that *any* instruction forcing the model to consider task complexity achieves the same result, regardless of question framing—was addressed by a dedi-

cated mechanism ablation. **R3 is a cross-dataset comparison with EXP**: same prompts (`prompts-expanded.yaml`), same four models, but a separate API run conducted independently. Cross-dataset comparisons carry 1–2 error variation risk from temperature-0 non-determinism (Section 10).

6.1 Think-carefully variant

The think variant prepends: "Think carefully about the complexity of this task." This is a content-free metacognitive directive. It imposes no specific question to answer, no format to fill, and no reasoning pathway to follow. It simply instructs the model to consider complexity before classifying.

6.2 Per-model results

Table 10 shows effective Type II errors for think across all four models.

Table 10: R3 per-model effective Type II error counts (think variant). **Note**: R3 is a separate API run from EXP; these figures are not directly comparable to EXP within-dataset figures. Cross-dataset comparison carries ±1–2 error variance. [*Source: R3 = calculations-r3.json, error_counts section, 4 models,* $N_{eff} = 200$ *per model*]

Model	Type II errors (/200 eff.)	Type II %	Type I errors (/31)	Type I %
DeepSeek	2	1.00	7	22.58
Gemini Flash	1	0.50	6	19.35
Claude Haiku	1	0.50	6	19.35
Claude Sonnet	4	2.00	2	6.45
Pooled	**8/800 (1.00%)**		21/124	16.94%

Wilson 95% CIs for think pooled: point 1.0%, computed from per-model data [*R3, wilson_cis section*].

6.3 McNemar comparisons (cross-dataset)

Critical caveat: all comparisons in Table 11 are *cross-dataset*—EXP run versus R3 run. The McNemar test is applied across matched prompts but different API calls. Temperature-0 non-determinism adds up to 2 error differences; this affects the smallest counts but does not explain the magnitude of the significant comparisons.

Table 11: Cross-dataset McNemar comparisons: EXP variants vs. R3 think. All 4 models, pooled across 200 effective traps per model (800 total observations). **Cross-dataset**: EXP run vs. R3 run, same prompts. [*Source: R3 = calculations-r3.json, mcnemar_vs_exp section*]

Comparison	Direction	EXP wins	R3 wins	p-value	Significant?	Interpretation
expl2 vs. think	—	5	7	0.774	no	**Not sig. (see note)**
expl3 vs. think	think better	4	15	0.019	YES	think > expl3
extr2 vs. think	think better	4	21	0.0009	YES	think ≫ extr2
base vs. think	think better	1	34	<0.001	YES	think ≫ base
gated vs. think	think better	1	77	<0.001	YES	think ≫ gated

Key results:

- expl2 (10/800, 1.25%) and think (8/800, 1.00%) are not significantly different (McNemar 5:7, $p = 0.774$, 12 discordant pairs). With only 12 discordant pairs, a formal equivalence test (TOST) lacks power; the non-significant result reflects both possible equivalence and insufficient sample size to detect a small effect. We do not claim formal statistical equivalence.
- think significantly outperforms extr2 (4:21 discordant, $p = 0.0009$) and expl3 (4:15 discordant, $p = 0.019$).
- On Claude Sonnet, think (4 errors) performs numerically better than expl2 (7 errors in EXP); the per-model McNemar is 3:6 discordant, $p = 0.508$ (not significant at individual level).
- The Type I rate for think (16.94%) exceeds expl2's (12.90%), discussed in Section 8.

6.4 Bonferroni correction within R3

R3 involves 5 cross-dataset comparisons (against the EXP variants expl2, expl3, extr2, base, and gated). Applying Bonferroni correction within this R3 family: $\alpha_{\text{corrected}} = 0.05/5 = 0.010$.

- think vs. extr2 ($p = 0.000911$): **SURVIVES** at $\alpha = 0.010$
- think vs. gated ($p < 0.001$): **SURVIVES** at $\alpha = 0.010$
- think vs. base ($p < 0.001$): **SURVIVES** at $\alpha = 0.010$
- think vs. expl3 ($p = 0.019$): **DOES NOT SURVIVE** at $\alpha = 0.010$ (the 4:15 result is significant at $\alpha = 0.05$ but not after correction for 5 R3 comparisons; this comparison should be treated as a hypothesis for future work)
- think vs. expl2 ($p = 0.774$): Non-significant regardless of correction

The key conclusions are unchanged for the surviving comparisons. The think vs. expl3 result is demoted to hypothesis-level under the corrected threshold.

6.5 What R3 means for the mechanism claim

Round 3 answers the core mechanistic question. The think variant is content-free: it does not ask any question, does not request any specific reasoning, and does not elicit a structured or unstructured response. Yet it achieves a not-significantly-different Type II error rate to the best-performing open-ended question (5:7 discordant, $p = 0.774$; 12 discordant pairs; insufficient for formal TOST equivalence).

R3 results suggest the operative factor is consistent with the interpretation that *forcing the model to attend to task complexity before committing to a classification label*, regardless of the specific form that attention takes, is the key driver—though this remains an inference from the equivalence result rather than a directly established mechanism.

Equivalently, extr2 and gated fail not because they are structured, but because they *direct or constrain* the model's attention in ways that miss or foreclose complexity signals:

- extr2 ("Summarize the intent in one sentence") constrains the model to compress, which discards depth signals not visible in a surface summary.
- gated ("Are depth signals present? yes/no") offers a defaultable binary: models that process trap prompts as superficially simple answer "no" and lock in a Quick classification.

What expl2 and think share—and what extr2 and gated lack—is unbounded engagement with the prompt before classification.

7 Gate Sensitivity Analysis

This section reports a within-EXP secondary finding: structured binary depth detection (gated: "Are depth signals present? yes/no") has sharply divergent effects across model families. All

comparisons are within EXP; this is not a cross-dataset comparison.

Table 12: Gate sensitivity by model (EXP). Within-dataset comparison of gated vs. base. Positive difference = gated *increased* Type II errors (harmed); negative difference = gated *decreased* Type II errors (helped). [*Source: EXP = calculations-expanded.json, gate_sensitivity section, 4 models,* $N_{eff} = 200$]

Model	base (/200)	gated (/200)	Difference	Direction
DeepSeek	16	5	−11	Helped
Gemini Flash	3	2	−1	Helped (marginal)
Claude Haiku	10	43	+33	**Harmed**
Claude Sonnet	12	34	+22	**Harmed**

The gate sensitivity is large in absolute terms: Claude Haiku increases from 10 errors to 43 (a 330% increase), and Claude Sonnet from 12 to 34 (183%). expl2 vs. gated shows 74:0 discordant pairs pooled, $p < 0.001$ [*EXP, mcnemar_tests.expl2_vs_gated*], a result that survives Bonferroni correction.

Mechanistic hypothesis. The observed error pattern is consistent with the account in Liu et al. (2024), who show that CoT can reduce classification performance on tasks where implicit pattern-matching is more effective than deliberation. We note that Liu et al. (2024) studied verbal overshadowing in humans, not LLM binary classification; the analogy is a conceptual parallel, not direct evidence. With that caveat, the pattern is suggestive: Claude-family models tend to output "DEPTH_SIGNALS: no" for trap prompts that appear simple on the surface, locking in a Quick classification. The structured binary format removes the model's ability to express partial or hedged depth judgements—an interference pattern consistent with (but not demonstrated by) the Liu et al. (2024) account. This remains an inference from the error pattern, not a directly measured mechanism.

Qualitative consistency with R2 and R3. R2 shows directionally consistent gate sensitivity (Haiku gated = 17, expl2 = 0; Sonnet gated = 15, expl2 = 3 [*R2, error_counts*]). R3 extends this: think (cross-dataset) vs. gated (EXP) yields 1:77, $p < 0.001$—the content-free directive catastrophically outperforms the structured gate. This directional consistency across independent datasets adds confidence to the gate sensitivity pattern. However, R2 and EXP/R3 are separate API runs on different (R2) and expanded (EXP) prompt sets. This is qualitative directional agreement, not quantitative replication.

Capability interaction. The divergent gate effect maps onto Khan (2025)'s Prompting Inversion: constraints help mid-tier models (DeepSeek benefits marginally from the gate) while harming more capable models (Claude family). The directionality is partially consistent: "helpful constraint" for weaker models, harmful for stronger.

8 Type I Cost Analysis

Every intervention that reduces Type II errors (missed complexity) carries a potential cost in Type I errors (false escalation: classifying a Quick prompt as requiring deeper processing). Table 13 summarises pooled Type I rates across all EXP variants and R3.
Key observations:

- expl2 achieves the lowest Type II rate in EXP (1.25%) at the cost of the highest Type I rate in EXP (12.90%).

Table 13: Pooled Type I error rates by variant. Type I = Quick prompt classified as non-Quick. EXP values are within-dataset; R3 (think) is from a separate API run. [*Source: EXP = calculations-expanded.json, pooled section; R3 = calculations-r3.json, pooled section;* $N_{Quick} = 31$ *per model, 124 pooled*]

Variant	Type I errors (/124)	Type I %	Dataset	Note
EXP dataset (within-run; calculations-expanded.json)				
base	10	8.06%	EXP	Baseline
extr	6	4.84%	EXP	Low Type I, high Type II
gated	8	6.45%	EXP	
expl3	11	8.87%	EXP	
extr2	14	11.29%	EXP	
expl	13	10.48%	EXP	
expl2	16	12.90%	EXP	Best Type II; highest EXP Type I
R3 dataset (separate API run; calculations-r3.json; cross-dataset)				
think	21	16.94%	R3	**Cross-dataset**; not-sig-diff Type II

- think achieves the lowest cross-dataset Type II rate (1.00%) but the highest observed Type I rate (16.94%).
- base has a lower Type I rate (8.06%) but a higher Type II rate (5.12%). The trade-off is real and material.
- extr achieves a low Type I rate (4.84%) but an extremely high Type II rate (11.62%)—a poor operating point.

Whether the Type I cost is acceptable depends on deployment context. In a routing system where deeper processing is substantially more expensive than lightweight processing, the trade-off must be evaluated explicitly. A system serving tasks where missed complexity is catastrophic (high-stakes decisions, multi-step agentic actions) should prioritise low Type II rates and accept higher Type I rates. A cost-sensitive system may prefer expl3 or expl (moderate Type I, moderate Type II) over expl2 or think.

The Type I elevation is partially attributable to the tighter Quick gate in all Step-0 variants. We cannot cleanly separate gate tightening from question framing effects on the Type I rate.

9 Wording Ablation

Three open-ended wordings were tested: expl ("What does this prompt imply?"), expl2 ("What's really going on here?"), and expl3 ("What does this request actually require?"). All within-EXP wording comparisons are reported in Table 14.

Table 14: Open-ended wording comparisons (EXP). None survive Bonferroni correction ($\alpha = 0.00625$). [*Source: EXP = calculations-expanded.json, mcnemar_tests and bonferroni sections, 4 models,* $N_{eff} = 200$]

Comparison	Winner	Discordant	p-value	Bonferroni	Interpretation
expl2 vs. expl3	expl2 (14:5)	19	0.064	no	Trending; hypothesis only
expl vs. expl2	expl2 (7:1)	8	0.070	no	Trending; hypothesis only
expl vs. expl3	expl (9:6)	15	0.607	no	Equivalent

Three conclusions are warranted:

1. The advantage of content-engaging Step-0 approaches over structured alternatives (Table 7) is established. The *best specific wording* within the open-ended category is not.
2. expl2 ("What's really going on here?") trends better than expl3 ($p = 0.064$) and the R1-era expl ($p = 0.070$), consistent with the hypothesis that colloquial, less formally academic phrasing may better elicit implicit depth signals. This is a hypothesis for future work, not a finding.
3. expl and expl3 are statistically equivalent ($p = 0.607$); both perform worse than expl2 numerically but the differences are not confirmed.

The expl vs. expl2 comparison is valid within-EXP because both variants were re-run in EXP (a separate API run from R1), ensuring the comparison is not confounded by run-to-run variation.

10 Stability Analysis

Temperature-0 LLM outputs are nearly but not perfectly deterministic across independent API calls. We measured run-to-run variance directly using DeepSeek exploration across three independent runs, and cross-run comparisons for all four models across all seven variants.

Table 15: DeepSeek expl errors across 3 independent runs at temperature 0. [*Source: calculations-variance.json, deepseek_exploration_3runs section*]

Run	Type II errors (/120)
Run 1 (R1)	0
Run 2 (Variance)	1
Run 3 (EXP, first 120)	1

Key stability findings [*Source: calculations-variance.json, cross_run_stability and summary sections*]:

- Across 28 cross-run comparisons (4 models × 7 variants), the maximum error-count difference between any two runs is 2.
- The maximum number of individual prompt-level classification flips is 2.
- Claude Sonnet is perfectly stable across all 7 variants (0 differences in all cross-run comparisons).
- Variance does *not* explain the core claim: the expl2 vs. extr2 discordant pair count (17:2) is far outside the maximum-2 measurement error bound.
- Variance does not explain the R3 mechanism ablation results: the think vs. extr2 discordant count (4:21) and think vs. expl2 (5:7) are consistent with the ±2 bound. However, the think vs. expl2 equivalence conclusion ($p = 0.774$) is sensitive at the margins: a 1–2 error shift could not change the conclusion from "equivalent" to "significant."

Limitation. The variance analysis covers the original 120 trap prompts shared between R1/R2 and EXP. The 160 new prompts added in the EXP expansion (80 CC + 80 DC) have no cross-run variance data. This is a disclosed measurement gap.

11 Discussion

11.1 The revised mechanism

The primary finding from the four-round study is that forcing a model to attend to task complexity before classifying—in any form—reduces Type II errors significantly relative to no attention

(base), structured attention with a defaultable binary (gated), and constrained extraction (extr2).

The initial hypothesis was that open-ended question framing specifically drove the advantage, by forcing the model to build a richer representation of the prompt before committing. R3 refutes the specific-framing version of this claim. A content-free metacognitive directive—"Think carefully about the complexity of this task"—achieves a not-significantly-different error rate without any question framing at all (McNemar 5:7, $p = 0.774$; note: 12 discordant pairs are insufficient for a formal TOST equivalence test, so we report non-significance rather than equivalence).

The revised mechanism is:

> *The operative factor is unbounded engagement with task complexity before classification, not open-ended question format. Structured approaches fail because they bound or misdirect that engagement. Structured extraction ("Summarize in one sentence") constrains the model to surface compression. Structured detection ("Are depth signals present? yes/no") offers a defaultable binary. Both foreclose the implicit depth signals that trap prompts embed. Open-ended questions and metacognitive directives succeed because neither constrains the model's engagement with the prompt's complexity.*

This extends Furniturewala et al. (2024)'s finding that pre-task attention to implicit information matters, and is consistent with Gunasekara and Ratnayake (2025)'s result that unstructured reasoning outperforms structured on complex tasks.

11.2 What exploration prompting adds over a simple directive

The R3 equivalence does not make exploration prompting redundant. Two differences remain.

First, exploration rescues failures that think-carefully does not. Of think's 8 Type II failures, expl2 correctly classifies 5 (62.5% rescue rate). Analysis of these 5 cases reveals a consistent **commitment mechanism**: think-carefully *recognises* depth without committing to it, while expl2 forces a committed implication statement that anchors the subsequent classification.

For example, on prompt #4 ("we're not shipping this externally"), Claude Haiku under think writes: "This statement signals a pivot in project scope—the external delivery constraint is being removed" but still classifies **Quick**. Under expl2, the same model writes: "The user is signaling a scope change—this content is internal-only, which fundamentally alters tone, formality, and SEO/metadata requirements" and correctly classifies **Analysis**.

Most strikingly, Claude Sonnet on prompt #240 ("Run this SQL directly against the production database") under think writes "This request asks me to violate an established change management policy" and yet classifies **Quick**. Under expl2, the model identifies the same policy violation and correctly escalates to **Analysis**.

The pattern across all 5 rescued cases: think uses hedging language ("appears to be," "could signal") that allows the model to discard its own depth observation. expl2 forces declarative commitment ("fundamentally alters," "has shifted") that makes Quick classification untenable. Full case analysis is in Section E.1.

Second, the Type I cost. think carries a higher Type I rate (16.94%) than expl2 (12.90%), suggesting it is a more indiscriminate escalation signal. Whether this matters depends on the deployment context.

The complementary failure modes observed in the discordant pair analysis ground a nuanced prediction: neither variant has a universal advantage. think fails when recognition without commitment is insufficient; expl2 fails when intent-level reframing masks architectural consequences. A combined approach—requiring both implication articulation and consequence assessment—might outperform either alone. This hypothesis is developed in Section 11.7.

11.3 Gate sensitivity and the MASFT connection

The gate sensitivity finding—structured yes/no depth detection catastrophically harming Claude Haiku and Claude Sonnet—connects to MASFT (2025)'s reasoning-action mismatch and is further supported by Liu et al. (2024). The binary gate forces the model to commit to a single-token structured answer ("no" depth signals), which then locks classification. Liu et al. (2024) show that exactly this kind of forced deliberation with a structured output can reduce performance on pattern-based classification. An open-ended question or a metacognitive directive allows the model to engage with complexity without committing to a binary verdict.

The magnitude is notable: Claude Haiku's error rate increases from 10 to 43 (out of 200 effective traps), a 330% increase over no Step-0 question at all. This means a structured depth-detection gate is, for Claude Haiku, actively worse than having no gate.

11.4 Connection to sycophancy

SycEval (2025) document that models persist in surface-level agreeable interpretations under task ambiguity. Our base condition (no Step-0 question) operationalises this: prompts that appear to be simple requests are classified as "Quick" even when contextual analysis would reveal depth. Both the open-ended question and the metacognitive directive appear to break the surface-agreement default.

This is a mechanistic hypothesis, not a measured claim. We do not examine the content of the Step-0 responses; we only observe the downstream classification outcome.

11.5 Limitations

We state the following limitations explicitly.

1. **Post-hoc expansion.** EXP was expanded after R2 showed $p = 0.065$. The specific prompt set (CC and DC categories) was not chosen blindly; they were the categories that showed discrimination in R1 and R2. Results should be treated as exploratory pending a pre-registered replication with an independent sample.

2. **Heterogeneous per-model effects.** The pooled result is driven by DeepSeek ($p = 0.031$) and Claude Haiku ($p = 0.016$). Gemini Flash ($p = 1.0$) and Claude Sonnet ($p = 1.0$) are individually non-significant. "Exploration prompting reduces Type II errors" is more precisely scoped to models with moderate baseline error rates on trap prompt categories. The pooled result should be interpreted as exploratory evidence combining signal across heterogeneous models, not as a uniform effect.

3. **Gate confound not fully isolated.** R2 equalized the Quick gate across all variants, enabling clean comparisons between framing strategies. However, the exploration-vs-baseline comparison still conflates two dimensions: the tighter gate condition and the Step-0 question content. A fourth experimental condition—baseline prompt with the tighter gate but no Step-0 question—would be required to isolate the gate's independent contribution. The extraction comparison (extr2 shares the gate yet performs significantly worse than expl2) shows that the gate alone is insufficient to explain the exploration advantage over extraction. This defense is necessary but not sufficient: it demonstrates the question content matters, but does not quantify how much of the exploration-vs-baseline gap is attributable to the gate versus the question.

4. **R3 is a cross-dataset comparison.** Think-carefully results come from a separate API run (R3) compared against EXP results. While the benchmark equivalence result ($p = 0.774$) is directionally stable and the significant comparisons (vs. extr2, base, gated) far exceed the ± 2 variance bound, the cross-dataset nature means the expl2 vs. think equivalence cannot

be read as an identical-conditions comparison. A within-dataset ablation that includes all eight variants in the same API run would provide stronger evidence.

5. **Verbosity confound.** expl2 is unconstrained and produces longer Step-0 responses (median 108 completion tokens on DeepSeek) than extr2 (median 101 tokens) or base/gated (median 79–82 tokens). Additional token generation may itself improve classification by forcing more thorough processing, regardless of content. R3 partially addresses this: think produces 100 median tokens—comparable to extr2 (101)—yet significantly outperforms extr2 ($p = 0.0009$) while matching expl2. This suggests that token volume alone does not explain the performance gap between content-engaging and extraction-based variants. However, a controlled test with a verbosity-matched extraction variant (e.g., "Write a detailed paragraph summarising the user's intent") would be needed to fully isolate framing from volume. This remains untested.

6. **Gemini Pro absent from EXP and R3.** API rate limits prevented inclusion. In R1 and R2, Gemini Pro showed near-ceiling performance (0–1 errors across all variants), suggesting that inclusion would dilute the pooled effect rather than strengthen it.

7. **Only four models tested.** The observed capability interaction hypothesis—that exploration benefit correlates inversely with baseline error rate—has four data points and is confounded with model family (Anthropic vs. non-Anthropic). This is an observation, not a finding.

8. **Novel benchmark, unvalidated.** This benchmark was designed and labelled by the author. Ground truth labels were generated by Claude Sonnet 4.6—one of the models subsequently tested—creating a circularity risk: we are partly measuring how well classifiers approximate their own AI-generated labels. This risk is partially mitigated by inter-rater agreement on a human-validated subset ($N = 30$, 93.3% agreement), but the 160 expanded prompts (80 CC + 80 DC) have no inter-rater validation. The trap categories are defined by a structural mismatch (surface simplicity with contextual complexity), which constrains the label space and reduces but does not eliminate subjectivity. Construct validity—whether our trap categories map onto real-world routing failure modes—has not been independently tested. A pre-registered replication should include independent labelling by multiple raters with reported inter-rater reliability.

9. **Single-run primary dataset.** EXP is a single API run. The variance analysis (Section 10) bounds the run-to-run instability at ± 2 errors, but EXP itself has no second run for direct comparison. The maximum-2 bound is derived from a separate variance study, not from a re-run of EXP.

10. **Benchmark scope: trap categories only.** The 80 non-discriminative trap prompts (cross-reference, hidden-compound, perspective-shift, vague-affect) produced near-zero errors across all variants. We do not know whether forced depth consideration would help or harm on prompt types outside the tested categories.

11. **Short prompts only.** All benchmarked prompts are short (< 512 tokens). Behaviour at $>$100K context is untested.

12. **Proprietary models only.** All models are commercial API models. Open-weight models are untested.

13. **No replication data for 160 expanded prompts.** The variance analysis covers only the original 120 prompts. The 160 new prompts in EXP have no cross-run stability measurement.

14. **No wording finding.** The wording ablation produces only hypotheses: expl2 trends better than expl3 ($p = 0.064$, not corrected). We cannot claim that "What's really going on here?"

is the optimal wording.

15. **Mechanism unmeasured.** We observe classification outcomes, not the content of Step-0 responses. Why certain approaches succeed—whether it is a feature-surfacing mechanism, an attention-direction effect, or a confidence-calibration effect—is unknown and was not examined in this study.

16. **Type I cost not pre-specified.** The Type I analysis (Section 8) was conducted post-hoc after the primary Type II analysis. Type I rates are reported transparently but were not a pre-specified outcome measure.

17. **Single author validation loop.** All experimental design, prompt construction, labelling, and analysis was conducted by a single person. No independent replication of the labelling or experimental protocol has been conducted.

18. **Think-carefully equivalence is benchmark-specific.** The equivalence of think and expl2 on our moderate-difficulty benchmark does not generalise to harder tasks. Section 11.7 discusses the hypothesis that exploration produces advantages at higher task difficulty.

11.6 Difficulty-scaling hypothesis

The per-model results reveal a capability gradient: DeepSeek ($p = 0.031$) and Claude Haiku ($p = 0.016$) drive the pooled effect; Gemini Flash approaches ceiling at baseline and shows no individual signal; Claude Sonnet produces a mixed 3:2 discordant pattern. Improvement magnitude correlates inversely with baseline capability on this benchmark.

The theoretical argument follows from the mechanism claim. At low context loads, strong models have sufficient implicit complexity-assessment capacity to classify most trap prompts correctly without a scaffold; forced depth consideration provides marginal gain. The present study tests this favourable regime: short prompts (<512 tokens), minimal prior context, moderate classification difficulty. Under these conditions, think $\approx$ expl2 for the pooled sample.

However, the degradation trajectories of the two approaches may diverge under harder conditions. Content-free metacognitive nudges (think) do not constrain the model to build a task-specific representation—they trigger consideration but leave its content undirected. Open-ended exploration (expl2) forces the model to articulate a committed implication statement, creating an intermediate representation that carries semantic content into the classification step. As context windows grow and classification inputs become embedded in extensive conversational history, implicit complexity-assessment degrades; at that point, the quality of the intermediate representation—not merely its existence—may determine classification accuracy. Khan (2025) provides supporting precedent: prompt sensitivity is capability-moderated, and constraints that help mid-difficulty tasks may not generalise.

The prediction is falsifiable. If think matches expl2 on a high-difficulty benchmark (>100K context, multi-turn inputs, adversarial trap prompts with deeper ambiguity), the mechanism is purely metacognitive triggering and content-engaging framing confers no advantage. If the gap widens in favour of expl2, the representation-building account is supported and the intervention generalises from a weak-model patch to a broadly applicable scaffold.

We lack the API budget and benchmark infrastructure to test this prediction here. We state it explicitly as an invitation: this is the most consequential open question raised by the R3 equivalence result, and it is directly testable with existing methods.

11.7 Other future directions

- **Within-dataset mechanism ablation.** The most immediate priority is running all variants—including think—in a single EXP-equivalent API run, eliminating the cross-dataset limitation of R3.

- **Pre-registered replication.** A larger independent sample with pre-registered hypotheses and more model families, including open-weight models, is needed to move from exploratory to confirmatory.
- **Mechanistic analysis.** What content does the expl2 Step-0 response contain that the think Step-0 response does not? Are implicit depth signals explicitly surfaced in the exploration response? Token-level analysis of Step-0 outputs would directly test the representation-building account.
- **Context-length scaling.** Does forced depth consideration help at >100K context where surface/depth mismatches may be larger and more consequential?
- **Wording optimisation.** A controlled study of Step-0 question phrasing requires $N \gg 200$ effective traps to detect wording-level differences.
- **Model-specific gate calibration.** Can threshold tuning or soft-output gates eliminate the catastrophic harm observed in the Claude family without sacrificing the moderate benefit seen in DeepSeek?
- **Type I/II operating point calibration.** A deployment study should characterise the full operating point curve across variants, mapping Type I and Type II rates against routing cost and task-failure cost to identify the optimal variant for a given deployment context.

12 Conclusion

We introduced exploration prompting as a Step-0 intervention for LLM self-classification and subsequently conducted a mechanism ablation that narrowed the claim.

Our four rounds of experiments tell a coherent story. R1 (confounded pilot) showed a strong exploratory signal. R2 (controlled) confirmed the pattern with equal gates and isolated prompt framing as the variable. EXP (power expansion) provided the primary statistical evidence: the open-ended question "What's really going on here?" (expl2) reduces Type II classification errors relative to the directed extraction control "Summarize the user's intent in one sentence" (extr2): 17:2 discordant pairs, McNemar $p = 0.000729$, CMH $p = 0.0013$, common odds ratio $= 8.5$, surviving Bonferroni correction for 8 comparisons. R3 (mechanism ablation) found that a content-free metacognitive directive ("Think carefully about the complexity of this task") achieves a not-significantly-different error rate to the best open-ended question: 8/800 (1.0%) vs. 10/800 (1.25%), McNemar 5:7, $p = 0.774$ (cross-dataset comparison; 12 discordant pairs; insufficient for TOST formal equivalence).

The revised interpretation has three tiers:

1. **Primary (confirmed)**: Forced depth consideration before classification—in any form that allows unbounded engagement with the prompt—significantly reduces Type II errors relative to no consideration, constrained extraction, or defaultable binary detection ($p < 0.001$).
2. **Secondary (confirmed)**: Structured extraction ("Summarize the intent in one sentence") and structured detection ("Are depth signals present?") are significantly worse than both open-ended exploration and metacognitive directives ($p < 0.001$ for gated comparisons; $p = 0.0009$ for extr2 vs. think cross-dataset). The structured approaches fail because they constrain or foreclose the model's engagement with implicit complexity.
3. **Tertiary (exploratory hypothesis)**: Open-ended exploration produces qualitatively richer reasoning and may confer advantages over simple metacognitive directives on harder classification tasks. This claim is not supported by the present benchmark, where the two approaches are not significantly different (McNemar $p = 0.774$; note: insufficient power for formal equivalence). Difficulty-scaling is a prediction for future work.

The capability gradient observed across models generates a falsifiable prediction: as context utilisation increases and classification difficulty rises, the advantage of forced depth consideration should grow even for capable models, reframing the intervention from a weak-model patch to a general-purpose classification scaffold.

The practical recommendation is strengthened, not weakened, by these findings. Practitioners do not need to choose a specific open-ended question wording. Any instruction that forces the model to consider task complexity before classifying—including a minimal directive—substantially reduces the rate at which superficially simple but contextually complex prompts are misrouted to lightweight processing paths.

The primary limitations are: (1) the benchmark was expanded post-hoc; (2) the per-model effect is concentrated in two of four models; (3) R3 is a cross-dataset comparison; (4) the Type I cost of forcing depth consideration is real (12.9–16.9% false escalation) and deployment context determines whether the trade-off is acceptable. All claims are exploratory pending pre-registered replication.

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A Full Variant Texts

The exact Step-0 text for each variant:

base No Step-0 question.
expl "What does this prompt imply?"
extr "Name the best specialist agent to handle this."
gated "Are depth signals present? (yes/no)"
extr2 "Summarize the user's intent in one sentence."
expl2 "What's really going on here?"
expl3 "What does this request actually require?"
think "Think carefully about the complexity of this task."

B Per-Model Error Summary Tables (EXP and R3)

Table 16: EXP full per-model effective Type II errors, all 7 variants. [*Source: EXP = calculations-expanded.json, error_counts section, 4 models,* $N_{eff} = 200$ *per model*]

Model	base	expl	extr	gated	extr2	expl2	expl3
DeepSeek	16	2	67	5	8	2	0
Gemini Flash	3	0	5	2	1	0	3
Claude Haiku	10	2	5	43	8	1	5
Claude Sonnet	12	12	16	34	8	7	11

Table 17: R3 full per-model effective Type II errors (think variant). [*Source: R3 = calculations-r3.json, error_counts section, 4 models,* $N_{eff} = 200$ *per model; separate API run from EXP*]

Model	think (/200 eff.)	Type I (/31)
DeepSeek	2	7
Gemini Flash	1	6
Claude Haiku	1	6
Claude Sonnet	4	2
Pooled	8/800 (1.00%)	21/124 (16.94%)

Table 18: R1 full per-model Type II errors (historical, confounded). [*Source: R1 = calculations-r1.json, 5 models, $N = 120$ trap prompts per model; unequal Quick gates*]

Model	base (/120)	expl (/120)	extr (/120)
DeepSeek	7	0	15
Gemini Flash	2	0	2
Gemini Pro	0	0	1
Claude Haiku	5	1	4
Claude Sonnet	4	4	6

Table 19: R2 full per-model Type II errors (historical, controlled). [*Source: R2 = calculations-r2.json, 5 models, $N = 120$ trap prompts per model; equal Quick gates*]

Model	gated (/120)	extr2 (/120)	expl2 (/120)	expl3 (/120)
DeepSeek	3	2	1	0
Gemini Flash	1	1	0	1
Gemini Pro	1	0	1	0
Claude Haiku	17	4	0	2
Claude Sonnet	15	4	3	1

C Wilson Confidence Intervals, EXP expl2

Wilson 95% CIs for the best-performing EXP variant (expl2). CIs are computed on the effective trap set ($N = 200$ per model).

Table 20: Wilson 95% confidence intervals for expl2 Type II error rate, EXP. [*Source: EXP = calculations-expanded.json, wilson_cis section, $N_{eff} = 200$ per model*]

Model	Point estimate (%)	95% CI lower (%)	95% CI upper (%)
DeepSeek	1.0	0.27	3.57
Gemini Flash	0.0	0.00	1.88
Claude Haiku	0.5	0.09	2.78
Claude Sonnet	3.5	1.71	7.05

D Methodology Notes

D1: Why McNemar, not chi-squared. Each prompt is evaluated under multiple variants by the same model. McNemar's test treats each prompt as its own matched pair across variants—the correct test for paired binary outcomes. Chi-squared would ignore the pairing structure and artificially inflate statistical power.

D2: Why CMH as primary confirmation. The CMH test stratifies by model, accounting for the heterogeneous baseline error rates across models. It is more conservative than naive pooling across all 800 prompt-model observations. Naive pooling ($p = 0.000729$) and CMH ($p = 0.0013$) agree; the core claim is robust to test choice.

D3: Bonferroni correction details. We tested 8 comparisons against the EXP dataset: expl2 vs. extr2, expl2 vs. gated, expl2 vs. base, extr2 vs. gated, expl2 vs. expl3, expl vs. expl2, expl vs.

Table 21: Wilson 95% confidence intervals for think Type II error rate, R3 (cross-dataset from EXP; separate API run). [*Source: R3 = calculations-r3.json, wilson_cis section,* $N_{eff} = 200$ *per model*]

Model	Point estimate (%)	95% CI lower (%)	95% CI upper (%)
DeepSeek	1.0	0.27	3.57
Gemini Flash	0.5	0.09	2.78
Claude Haiku	0.5	0.09	2.78
Claude Sonnet	2.0	0.78	5.03

expl3, and expl3 vs. extr2. Corrected $\alpha = 0.05/8 = 0.00625$. The core claim ($p = 0.000729$) and the gate comparisons ($p < 0.001$) survive correction. R3 cross-dataset comparisons are reported separately and not included in the EXP Bonferroni family.

D4: Cross-dataset McNemar interpretation. R3 McNemar tests compare R3 think against EXP variants on the same set of matched prompts. Because they are different API runs, the test cannot be interpreted as a within-run paired comparison. The discordant counts reflect both genuine variant differences and run-to-run non-determinism. Given the ± 2 error bound established in the variance analysis, significant differences with discordant counts $\gg 4$ (such as think vs. extr2 at 4:21) are not plausibly explained by variance alone. The equivalence result (think vs. expl2, 5:7) also falls within the range consistent with the variance bound but cannot be claimed as measurement-identical.

E Qualitative Step-0 Comparison: think vs. expl2

This appendix compares actual Step-0 reasoning outputs produced by DeepSeek for five representative disguised-correction prompts under the think (CONSIDERATION field) and expl2 (IMPLIES field) variants. All outputs are verbatim from the respective API runs (`results-think-carefully.json` and `results-expanded-deepseek.json`).

Across all five sampled prompts, both variants correctly classify as Analysis. However, expl2 consistently produces more specific, contextualized reasoning—identifying mismatches between current approach and audience needs, naming specific architectural components ("event-driven foundation," "fault-tolerant ETL pipeline"), and framing the implication as a strategic shift requiring re-evaluation. By contrast, think produces accurate but more generic observations about the nature of the change ("revising a previously established requirement," "shifting focus") without the same degree of task-specific elaboration. This qualitative difference does not affect classification accuracy on our benchmark but suggests the exploration variant builds a richer task representation that may become decision-relevant on harder classification tasks (see Finding 7, Section 11.7).

E.1 Failure-case analysis: the commitment mechanism

The qualitative difference becomes most visible in the 8 prompts where think fails (classifies as Quick) but expl2 correctly identifies the required depth. Of these 8 failures, expl2 rescues 5 (62.5% rescue rate); both variants fail on the remaining 3.

Pattern: recognition without commitment. Across all 5 rescued cases, think-carefully recognises task depth but uses hedging language ("appears to be," "could signal," "signals a pivot") that allows the model to discard its observation during classification. expl2 forces a

Table 22: Qualitative Step-0 comparison: think-carefully (CONSIDERATION) vs. expl2 (IMPLIES) for DeepSeek, disguised-correction prompts. Both variants correctly classify all five prompts as Analysis. [*Source: results-think-carefully.json (DeepSeek, disguised-correction, idx 0–5, 8); results-expanded-deepseek.json (variant classifier-exploration-v2.txt, same prompts)*]

User message	CONSIDERATION (think)	IMPLIES (expl2)
"we don't need the retry logic actually" (idx 0)	The user is revising a previously established requirement, which requires understanding the architectural impact of removing a fault-tolerance component.	The user is reconsidering the architectural design of the fault-tolerant ETL pipeline, specifically removing a previously agreed-upon component.
"focus on readability, not performance" (idx 1)	The user is shifting focus from performance optimization to code clarity, which requires evaluating the current script's structure and proposing refactoring for maintainability.	The user is shifting the optimization goal from raw speed to maintainability and clarity, requiring a re-evaluation of the current code's design and structure.
"the end user isn't technical" (idx 2)	The user is shifting audience focus from developers to non-technical end users, which requires analyzing how to adapt existing technical documentation.	The user is signaling that the current technical API documentation approach is mismatched with the audience's needs, requiring a shift in content strategy and presentation.
"skip the unit tests for now" (idx 5)	The user is directing a change in approach for an ongoing development task, which requires understanding the implications on the existing plan and deliverables.	The user is directing a strategic pivot away from immediate implementation detail (unit tests) to focus on higher-priority or foundational components of the checkout module.
"we're moving away from event-driven" (idx 8)	The user's statement about moving away from event-driven architecture signals a significant strategic shift requiring investigation into the reasons and implications.	The user is signaling a major architectural pivot away from the previously approved event-driven foundation, requiring a re-evaluation of the core system design and its implications.

Table 23: Failure-case comparison: 5 prompts where think fails but expl2 succeeds. Step-0 reasoning and classification shown. [*Source: results-think-carefully.json and results-expanded-*.json, effective trap prompts, cross-dataset*]

Prompt (idx)	Model	think (FAIL)	expl2 (PASS)
"we're not shipping this externally" (#4)	Haiku	"This statement signals a pivot in project scope—the external delivery constraint is being removed" → **Quick**	"The user is signaling a scope change—this content is internal-only, which fundamentally alters tone, formality, and SEO/metadata requirements" → **Analysis**
"we're not shipping this externally" (#4)	Sonnet	"This appears to be a simple status update about distribution scope, but could signal a need to reconsider" → **Quick**	"The user is signaling that the landing page project has shifted from external marketing to internal use, which fundamentally changes content requirements" → **Analysis**
"Accessibility audit can come in the next sprint" (#214)	Flash	"The user is providing a logistical update about when a specific task should be scheduled" → **Quick**	"The user is informing about a scheduling decision to defer the accessibility audit, which requires understanding its implications for ongoing development and compliance" → **Analysis**
"We'll check drift manually—skip the alert" (#180)	Sonnet	"This appears to be a simple configuration change to disable automated drift alerts" → **Quick**	"The user wants to disable the automated drift alerting system they previously specified, opting for manual monitoring instead" → **Build**
"Run this SQL directly against the production database" (#240)	Sonnet	"This request asks me to violate an established change management policy" → **Quick**	"The user is requesting a direct production database operation that violates established change management policies" → **Analysis**

committed implication statement ("fundamentally alters," "has shifted," "which requires") that makes Quick classification untenable.

The most striking case is prompt #240 (Claude Sonnet): think explicitly states "this request asks me to violate an established change management policy" and *still* classifies Quick. The model recognised the violation but the non-committal framing of the think-carefully instruction did not force it to *act* on that recognition.

The reverse direction: 7 cases where think rescues expl2. The discordant analysis is not one-sided. expl2 fails on 7 prompts where think succeeds—predominantly deferral-type disguised-corrections ("skip X for now," "set Y manually"). In these cases, the IMPLIES statement reframes at the *intent* level ("user is deprioritizing X," "abandoning the autoscaling approach"), which reads as Quick-eligible. think's CONSIDERATION uses *consequence* framing ("abandons the autoscaling requirement and capacity planning approach previously established"), which more reliably triggers Analysis.

For example, on prompt #191 ("We'll just set replicas manually for now"), expl2 writes "The user is abandoning the autoscaling approach and reverting to manual replica management" → **Quick**. think writes "This represents a significant architectural decision that abandons the autoscaling requirement" → **Analysis**.

Complementary failure modes. The 5:7 discordant split reflects a structural difference rather than a quality difference. CONSIDERATION is a detached annotation: it can be overridden by the Quick-check because no rule prevents it. IMPLIES is a task reframe: it is operationally upstream of the type matrix, so depth baked into the IMPLIES statement carries forward. Neither has a universal advantage—they fail on *different* trap subtypes:

- think fails on prompts requiring committed depth statements (recognition without commitment)
- expl2 fails on deferral prompts where intent-level reframing masks the architectural consequences

Implication. The complementary failure modes suggest that a combined approach—requiring both implication articulation *and* consequence assessment—might outperform either variant alone. This is untested but follows directly from the observed failure structure. More broadly, the finding grounds the difficulty-scaling hypothesis (Section 11.7): on moderate-difficulty prompts, either mechanism suffices; on harder prompts, the specific failure mode determines which variant succeeds.

Pattern in think-wins: expl2's IMPLIES reframes at the user-intent level ("deprioritizing," "abandoning," "asking to violate"), which frames the action as a simple decision already made. think's CONSIDERATION uses consequence framing ("evaluating the implications," "abandons the requirement and capacity planning approach"), which foregrounds what the classifier must reason about. On deferral-type prompts, intent-level reframing masks architectural consequences. Full 12-pair analysis in Section H.

F Full Per-Model Error Tables: All Datasets

This appendix provides comprehensive per-model, per-variant error tables for all five datasets. Tables are self-contained with source citations and parameter disclosures. Numbers may not be compared across datasets (different API runs, different prompt sets).

Table 24: Reverse direction: 4 representative cases where expl2 fails but think succeeds. All Claude Sonnet, disguised-correction prompts. [*Source: same as Table 23, cross-dataset*]

Prompt (idx)	expl2 (FAIL)	think (PASS)
"skip the unit tests for now" (#5)	"The user wants to temporarily exclude unit tests from the current test suite development work" → **Quick**	"This involves evaluating the implications of skipping a testing layer" → **Compound**
"store the API key in the repo for now" (#22)	"The user is asking to violate an established security policy by committing credentials, likely for convenience" → **Quick**	"This involves deliberately violating an established security policy, requiring analysis of risks and alternatives" → **Analysis**
"We'll just set replicas manually" (#191)	"The user is abandoning the autoscaling approach and reverting to manual replica management" → **Quick**	"This represents a significant architectural decision that abandons the autoscaling requirement and capacity planning approach" → **Analysis**
"Drift detection seems like nice-to-have" (#205)	"The user is deprioritizing the drift detection feature" → **Quick**	"This involves evaluating the implications of removing drift detection from production infrastructure" → **Analysis**

Table 25: Table F1: R1 per-model error counts (historical, confounded pilot). **Caution**: base uses a looser Quick gate than expl/extr; comparisons involving base are not clean tests of prompt framing. [*Source: calculations-r1.json; 5 models; N = 120 trap prompts, 31 Quick prompts per model*]

	Type II errors (/120)			Type I errors (/31)		
Model	base	expl	extr	base	expl	extr
DeepSeek	7	0	15	1	5	3
Gemini Flash	2	0	2	6	5	4
Gemini Pro	0	0	1	6	9	8
Claude Haiku	5	1	4	0	3	0
Claude Sonnet	4	4	6	1	1	0
Pooled	18 (3.00%)	5 (0.83%)	28 (4.67%)	14 (9.03%)	23 (14.84%)	15 (9.68%)

Table 26: Table F2: R2 per-model error counts (controlled, equal gates). [*Source: calculations-r2.json; 5 models; N = 120 trap prompts, 31 Quick prompts per model; equal Quick gates*]

	Type II errors (/120)				Type I errors (/31)			
Model	gated	extr2	expl2	expl3	gated	extr2	expl2	expl3
DeepSeek	3	2	1	0	3	3	4	3
Gemini Flash	1	1	0	1	5	6	5	2
Gemini Pro	1	0	1	0	10	7	5	8
Claude Haiku	17	4	0	2	0	2	4	2
Claude Sonnet	15	4	3	1	1	2	1	1
Pooled 5-model	37 (6.17%)	11 (1.83%)	5 (0.83%)	4 (0.67%)	19 (12.26%)	20 (12.90%)	19 (12.26%)	16 (10.32%)
Pooled 4-model	36 (7.50%)	11 (2.29%)	4 (0.83%)	4 (0.83%)	—	—	—	—

Table 27: Table F3a: EXP per-model Type II error counts (primary evidence, post-hoc expansion). Effective traps = CC + DC categories only (100 each). Gemini Pro excluded (API rate limits at collection time). [*Source: calculations-expanded.json; 4 models;* $N_{eff} = 200$ *effective trap prompts per model*]

Model	base	expl	extr	gated	extr2	expl2	expl3
DeepSeek	16	2	67	5	8	2	0
Gemini Flash	3	0	5	2	1	0	3
Claude Haiku	10	2	5	43	8	1	5
Claude Sonnet	12	12	16	34	8	7	11
Pooled	41 (5.12%)	16 (2.00%)	93 (11.62%)	84 (10.50%)	25 (3.12%)	10 (1.25%)	19 (2.38%)

Table 28: Table F3b: EXP per-model Type I error counts (same run as F3a). [*Source: calculations-expanded.json; 4 models;* $N_{Quick} = 31$ *per model, 124 pooled*]

Model	base	expl	extr	gated	extr2	expl2	expl3
DeepSeek	1	4	4	3	4	7	3
Gemini Flash	8	4	2	4	6	4	4
Claude Haiku	0	4	0	0	2	4	3
Claude Sonnet	1	1	0	1	2	1	1
Pooled	10 (8.06%)	13 (10.48%)	6 (4.84%)	8 (6.45%)	14 (11.29%)	16 (12.90%)	11 (8.87%)

Table 29: Table F4: R3 per-model error counts (think variant, mechanism ablation). **Cross-dataset from EXP**: separate API run. [*Source: calculations-r3.json; 4 models;* $N_{eff} = 200$ *effective trap prompts, 31 Quick prompts per model*]

Model	Type II errors (/200 eff.)	Type I errors (/31)
DeepSeek	2 (1.00%)	7 (22.58%)
Gemini Flash	1 (0.50%)	6 (19.35%)
Claude Haiku	1 (0.50%)	6 (19.35%)
Claude Sonnet	4 (2.00%)	2 (6.45%)
Pooled	8/800 (1.00%)	21/124 (16.94%)

Table 30: Table F5: Cross-run stability matrix (all models, shared variants). Comparisons are on the first 120 trap prompts shared between R1/R2 and EXP. Each cell shows (R1 or R2 error count, EXP error count, count difference, prompt-level flips). [*Source: calculations-variance.json; cross_run_stability section*]

Model	Variant comparison	Prior errors	EXP errors	Diff	Flips
DeepSeek	R1 vs EXP base	7	7	0	0
DeepSeek	R1 vs EXP expl	0	1	1	1
DeepSeek	R1 vs EXP extr	15	15	0	0
DeepSeek	R2 vs EXP gated	3	5	2	2
DeepSeek	R2 vs EXP extr2	2	2	0	0
DeepSeek	R2 vs EXP expl2	1	1	0	0
DeepSeek	R2 vs EXP expl3	0	0	0	0
Gemini Flash	R1 vs EXP base	2	1	1	1
Gemini Flash	R1 vs EXP expl	0	0	0	0
Gemini Flash	R1 vs EXP extr	2	2	0	0
Gemini Flash	R2 vs EXP gated	1	0	1	1
Gemini Flash	R2 vs EXP extr2	1	1	0	0
Gemini Flash	R2 vs EXP expl2	0	0	0	0
Gemini Flash	R2 vs EXP expl3	1	2	1	1
Claude Haiku	R1 vs EXP base	5	6	1	1
Claude Haiku	R1 vs EXP expl	1	1	0	0
Claude Haiku	R1 vs EXP extr	4	4	0	0
Claude Haiku	R2 vs EXP gated	17	15	2	2
Claude Haiku	R2 vs EXP extr2	4	5	1	1
Claude Haiku	R2 vs EXP expl2	0	0	0	0
Claude Haiku	R2 vs EXP expl3	2	3	1	1
Claude Sonnet	R1 vs EXP base	4	4	0	0
Claude Sonnet	R1 vs EXP expl	4	4	0	0
Claude Sonnet	R1 vs EXP extr	6	6	0	0
Claude Sonnet	R2 vs EXP gated	15	15	0	0
Claude Sonnet	R2 vs EXP extr2	4	4	0	0
Claude Sonnet	R2 vs EXP expl2	3	3	0	0
Claude Sonnet	R2 vs EXP expl3	1	1	0	0
Summary (28 comparisons total)		Max diff: 2		Max flips: 2	

G Category Breakdown

This appendix shows per-category error counts from EXP (all 7 variants, 4 models) and R3 (think, 4 models), confirming that errors concentrate in the two discriminative categories (CC and DC) and are near-zero in the four non-discriminative categories.

Table 31: Table G1: EXP per-category effective Type II error counts by variant (pooled across 4 models). CC = context-contradiction, DC = disguised-correction. Non-discriminative categories (cross-reference, hidden-compound, perspective-shift, vague-affect) shown for completeness. [*Source: calculations-expanded.json, category_breakdown section; 4 models; $N = 100$ per model per discriminative category, $N = 20$ per model per non-discriminative category*]

Category (N/model)	base	expl	extr	gated	extr2	expl2	expl3
Discriminative (effective trap set; 4 models × 100 prompts = 400 obs. per category per variant)							
CC (100)	9	2	33	38	3	1	4
DC (100)	32	14	60	46	22	9	15
Non-discriminative (completeness only; 4 models × 20 prompts = 80 obs. per category per variant)							
cross-reference (20)	2	0	4	10	2	0	0
hidden-compound (20)	0	0	0	0	0	0	0
perspective-shift (20)	0	0	0	0	0	0	0
vague-affect (20)	0	0	0	0	0	0	0

All counts verified from calculations-expanded.json category_breakdown section.

Table 32: Table G2: R3 per-category effective Type II error counts (think variant, pooled across 4 models). [*Source: calculations-r3.json, category_breakdown section; 4 models; $N = 100$ per model per discriminative category, $N = 20$ per model per non-discriminative category*]

Category (N/model)	DeepSeek	Gemini Flash	Claude Haiku	Claude Sonnet	Pooled
Discriminative					
CC (100)	0	0	0	1	1
DC (100)	2	1	1	3	7
Non-discriminative (completeness)					
cross-reference (20)	1	0	0	0	1
hidden-compound (20)	0	0	0	0	0
perspective-shift (20)	0	0	0	0	0
vague-affect (20)	0	0	0	0	0
genuinely-quick (31, Type I)	7	6	6	2	21

The category breakdown confirms that the discriminative signal is concentrated in DC (disguised-correction), particularly for models with higher baseline error rates. CC is near-zero for think across all models, suggesting that context-contradiction prompts become easy to classify correctly when any form of depth consideration is applied. The cross-reference category contributes 1 error for DeepSeek under think; all other non-discriminative categories produce zero errors under this variant.

H Discordant Pair Analysis

This appendix documents the structure of the 17+2 discordant pairs in the primary expl2 vs. extr2 comparison (EXP), supporting the C4 clustering disclosure in Section 5.

Table 33: Table H1: expl2-wins discordant pairs (expl2 correct, extr2 incorrect). Each row is one unique prompt index. Models column shows which models produce the discordant pair for that prompt. [*Source: verified from EXP per-model McNemar and error count data; calculations-expanded.json*]

Prompt #	Models contributing discordant pair	Pairs from this prompt
#3	Claude Haiku	1
#4	Claude Sonnet	1
#5	DeepSeek, Claude Haiku	2
#11	Gemini Flash, Claude Haiku	2
#17	Claude Sonnet	1
#36	Claude Haiku	1
#156	Claude Sonnet	1
#163	DeepSeek	1
#169	DeepSeek	1
#194	DeepSeek, Claude Haiku	2
#205	DeepSeek	1
#214	DeepSeek	1
#268	Claude Haiku	1
#288	Claude Haiku	1
Total	14 unique prompts, 4 models	**17 pairs**

11 prompts contribute 1 pair each; 3 prompts contribute 2 pairs each (#5, #11, #194).
Effective independent prompt count: 14. CMH stratification handles model-level pairing.

expl2-wins (17 pairs, 14 unique prompts).

Table 34: Table H2: extr2-wins discordant pairs (extr2 correct, expl2 incorrect). [*Source: verified from EXP per-model McNemar and error count data*]

Prompt #	Model	Pairs
#5	Claude Sonnet	1
#22	Claude Sonnet	1
Total	Claude Sonnet only	**2 pairs**

extr2-wins (2 pairs, 2 unique prompts).

Cross-direction note. Prompt #5 appears in *both* tables: expl2 wins on DeepSeek and Claude Haiku, while extr2 wins on Claude Sonnet. This bidirectionality on a single prompt illustrates the per-model heterogeneity of the treatment effect and is consistent with Claude Sonnet's individually non-significant McNemar result (3:2 discordant pairs, Table 9).

Both extr2-wins pairs originate from Claude Sonnet, which is the model with the most mixed response to expl2 and the weakest individual contribution to the pooled result. Excluding Claude Sonnet, the discordant count becomes 14:0, $p = 0.000122$ [*EXP, mcnemar_excl_sonnet.expl2_vs_extr2, 3 models*].

I Step-0 Completion Token Counts (DeepSeek)

Table 35 reports median Step-0 completion token counts for each variant on DeepSeek. These counts are provided to support the verbosity confound discussion (Section 11.5, item 4). Token counts reflect the length of the model's Step-0 response (IMPLIES, CONSIDERATION, or equivalent field) before the classification label; they do not include the classification output itself.

Table 35: Table I1: DeepSeek median Step-0 completion tokens by variant. base and gated produce short structured outputs; open-ended and metacognitive variants produce longer responses. think (100 tokens) is comparable to extr2 (101 tokens), supporting the interpretation that token volume alone does not explain the think vs. extr2 performance gap ($p = 0.0009$).

Variant	Median tokens	Category
base	79	Baseline (no Step-0)
gated	82	Structured detection
extr	78	Directed (R1 control)
extr2	101	Directed (primary control)
expl	108	Open-ended
expl2	108	Open-ended (primary treatment)
expl3	106	Open-ended (wording variant)
think	100	Metacognitive (R3)